Non-Invasive Anemia Detection

Estimating haemoglobin from photographs of the palpebral conjunctiva

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Abstract — Screening for anaemia from smartphone photographs of the palpebral conjunctiva needs two components: an extractor that isolates the tissue, and a regressor that maps it to haemoglobin. This report covers the extractor. A learned U-Net, trained on 648 annotated images, captures haemoglobin-related colour far better than a classical rule-based extractor on 243 patients with laboratory haemoglobin (R^2 0.354 against 0.074 on 116 photographs it never trained on, read through a deliberately simple colour probe). Evaluated as segmentation models on 129 held-out subjects, an Attention U-Net improves on the U-Net (Dice 0.848 against 0.837; paired difference +0.010, $p = 0.007$) and cannot be separated from UNet++. Gray-world white balance, applied since the start, lowers segmentation accuracy, and serving the segmenter balanced images it was never trained on cost more (0.012 Dice) than the change of architecture gained. Two findings bound what this means for deployment: colour measured by one camera does not transfer to another, and a third public dataset of 26 children shows no colour signal, in part because how far the eyelid is pulled down changes conjunctival colour. The limiting factor is now the capture protocol and the deployment camera, not extraction.

Keywords — anaemia screening; palpebral conjunctiva; haemoglobin estimation; image segmentation; U-Net; Attention U-Net; white balance; domain shift; capture protocol

1. Summary

Scope. The project is divided into two workstreams: the extractor, which finds the conjunctiva and returns a clean crop, and the haemoglobin regressor (a ResNet trained on that crop) together with the mobile application. This report covers the extractor and the evaluation methodology; the regressor and the application are developed in parallel and are not reported here.

This period. Three encoder-decoder architectures — U-Net, UNet++ and Attention U-Net — were compared as segmentation models, with the metrics such models are judged by (Dice, IoU, precision, recall, boundary distance and the failure distribution) on 129 subjects held out from both training and model selection. Attention U-Net is the only one whose advantage over the U-Net resolves. Investigating white balance exposed a defect: the segmenter was trained on raw pixels but served white-balanced ones, which cost more accuracy than the architecture change gained; serving now matches training. Two production extractors, identical except for white balance, were trained on all 648 images so that the regressor can settle whether white balance affects haemoglobin error. Section 8.

Previous period. The learned extractor was compared with the classical one on **243 patients** rather than 26, using a public cohort of 217 subjects with laboratory haemoglobin. Read through a simple colour probe, it explains roughly three times the variance (R^2 0.354 against 0.074 on 116 photographs it never trained on). The same probe established that colour does not transfer between cameras, and that a third public dataset of 26 children does not show the colour signal at all, traced in part to how far the eyelid was pulled down. Section 7, which supersedes Section 6 on extraction.

The first period was spent building the image-processing and learning pipeline end to end: data loading and clinical labelling, conjunctiva extraction, quality control, the regression scaffold, an evaluation methodology, and a serving API. The guiding constraints are an eventually deployed screening app and small clinical cohorts — both punish silent failure, so the pipeline is engineered to produce numbers that can be defended: patient-grouped cross-validation, out-of-fold reporting only, and every result compared against a predict-the-mean baseline.

A publicly available dataset served as the first real test set: 52 photographs of 26 children, both eyes, with laboratory haemoglobin and a full blood count. Screening it produced two findings. The pipeline processes real photographs without error and 94% pass quality control, but a plain colour-threshold extraction — which scores almost perfectly on synthetic images — proved **unreliable on real tissue**; a refined extractor was then built against these captures and now produces clean masks across all 52. Separately, the per-patient WHO threshold was validated against real laboratory values: a fixed single cutoff would have missed 15% of the anemic children in this cohort. Both are reported in Section 4.

The principal work of this period was a **head-to-head comparison of the two candidate pipelines**, since one must eventually be deployed. Both were trained and scored on identical patients under an identical protocol, across five feature representations and the full metric suite (Section 6). Pipeline A leads B on 10 of 11 metrics fitted and 9 of 11 held out, and on every parameter-matched framing, but a paired test shows the two are **not statistically separable at 26 patients** ($p = 0.408$). The model **deployed at that stage** — pipeline A's erythema representation with the quality gate applied before fitting — reaches in-sample MAE 0.918 g/dL and R^2 +0.195, and on unseen patients MAE 1.033 and R^2 **+0.007** against a

leave-one-out baseline of 1.083 and −0.082: the first configuration here to beat its baseline on patients it has not seen, though a permutation test still puts it short of significance ($p = 0.067$). Its coefficients also pass a physiological check that the earlier channel-mean model failed (Section 6.9); it has since been replaced (Section 7.5). The regression network has not been trained on real photographs, so **no accuracy figure in this report is a clinical result**.

1.1 Abbreviations

Defined here on first use; the full glossary is in `GLOSSARY.md`.

Term	Expansion	Term	Expansion
Hb	Haemoglobin (g/dL)	MAE	Mean Absolute Error
WHO	World Health Organization	MSE / RMSE	Mean Squared Error / its root
QC	Quality Control	R^2	Coefficient of determination
ROI	Region Of Interest	LoA	Limits of Agreement (Bland–Altman)
CIELAB	CIE L*a*b* colour space	TP / FP / TN / FN	True/False Positive/Negative
L* / a* / b*	Lightness / green–red / blue–yellow axes	PPV / NPV	Positive / Negative Predictive Value
CLAHE	Contrast Limited Adaptive Histogram Equalisation	F1	Harmonic mean of precision and recall
RGB / sRGB / P3	Colour encodings and profiles	Dice / IoU	Segmentation overlap measures
CNN	Convolutional Neural Network	CV / LOO	Cross-Validation / Leave-One-Out
ResNet	Residual Network (CNN architecture)	OLS	Ordinary Least Squares
CBC	Complete Blood Count	L1 / L2	Absolute / squared coefficient penalties
U-Net	Encoder–decoder segmentation network	SD	Standard Deviation
HD95	95th-percentile boundary distance (pixels)	gw	Gray-world white balance

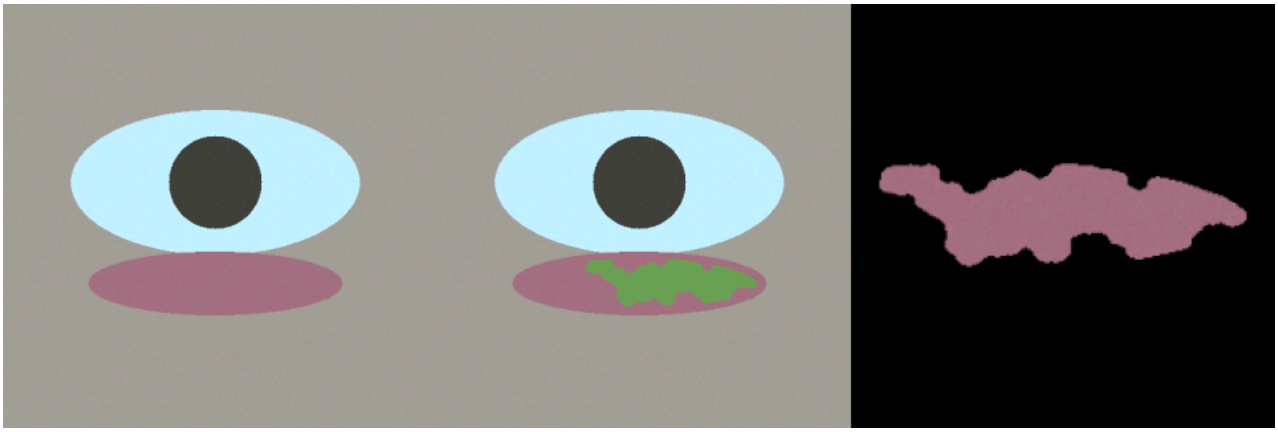
2. Pipeline design

The pipeline is a package of focused modules. The decision that matters most architecturally is that the inference path shares its preprocessing code with training but imports none of the training stack — so the deployed server cannot drift from the conditions the model was trained under, and does not load training machinery to answer a request.

Area	Design
Extraction	Built around a redness prior in CIELAB refined by a seeded grabCut; all backends share one interface so a trained model swaps in without touching other stages
Quality control	Enforced, not merely recorded: failing captures are dropped from training; the API returns a retake prompt rather than a number
Clinical labelling	WHO thresholds applied per patient from age and sex
Evaluation	Patient-grouped, haemoglobin-stratified cross-validation; out-of-fold predictions only; always reported against a predict-the-mean baseline
Checkpoints	Self-describing bundle carrying weights, standardisation constants, preprocessing settings and metrics
Performance	Preprocessing cached to disk, so segmentation runs once rather than once per epoch
Data handling	Adapters for both public datasets behind a common record type; adding the hospital data requires one adapter
Deployment	FastAPI service loading one model at startup
Testing	19 automated tests, plus synthetic eye phantoms allowing development without data

3. Choosing the extraction target: a controlled comparison

Two candidate colour rules were scored against known ground truth on synthetic phantoms. A brightness-based mask (bright, low-chroma pixels) scores a Dice coefficient of **0.000** — it selects the sclera and has no overlap with the conjunctiva. The redness-based extractor scores **0.999**. This is what fixed redness, not brightness, as the extraction target.



Left: input capture. Centre: region identified by the redness-based extractor, tinted green. Right: the square, undistorted crop passed to the regression network.

> **These are synthetic phantoms with flat colours and artificially clean separation.** A near-perfect score is expected and carries no clinical meaning. What the comparison establishes is that brightness-based masking targets the wrong tissue and redness-based extraction targets the right one. Section 4 shows what happens on real photographs.

4. Evaluation on real photographs

The first real test set is a publicly available dataset of 26 children, both eyes photographed, giving 52 close-up captures of the everted lower eyelid, accompanied by laboratory haemoglobin, date of birth, gender, and a full blood count. The captures are well framed and in focus — tighter on the conjunctiva than typical field photography. The dataset was not collected for this project; its source is to be cited (Section 10).

4.1 Cohort

Property	Value
Patients / images	26 / 52 (both eyes, exactly two captures each)
Haemoglobin	8.1 – 14.3 g/dL (mean 11.2, sd 1.37)
Anemic by WHO threshold	13 of 26 (50%)
Severe (Hb < 9.0)	3 patients
Age	3.4 – 16.7 years
Additional labels	Height, weight, socio-economic status, HCT, RBC, MCV, MCH, MCHC, RDW, platelets, MPV, TLC

Two properties make this cohort more useful than its size suggests. The 50/50 anemic split is unusually balanced — most cohorts skew heavily normal, which is what makes a predict-the-mean baseline hard to beat. And because both eyes of one child are photographed, the two captures are highly correlated: the loader assigns them a shared patient identifier so the grouped splitter cannot separate them; under a naive image-level split this dataset would produce substantially inflated results.

4.2 The pipeline runs on real data

Measure	Result
Images processed without error	52 of 52
Passed quality control	49 of 52 (94%)
Focus (Laplacian variance)	median 85.3, range 25.4 – 294.2
Mask coverage of frame	median 0.211, range 0.041 – 0.449
Rejections	3, all blur (25.4, 34.2, 34.9 against a threshold of 35.0)

Two of the three rejections sit marginally below the threshold on captures that look usable to the eye. The blur threshold was chosen on synthetic images and appears slightly too aggressive for real photographs; it should be recalibrated once ground-truth masks make the downstream effect measurable.

4.3 Extraction: failure found, then fixed

Quality-control statistics say whether a capture is usable, not whether the segmentation found the right structure. Visual inspection of the overlays answers that, and the answer is unfavourable: on a substantial fraction of the 52 real captures the colour prior bleeds onto the eyelashes and lower-lid skin, and occasionally onto the sclera. On several it extends well below the lash line onto the cheek.

The cause is straightforward: eyelashes and peri-orbital skin are reddish-brown enough to pass a redness test, and neither is dark enough in absolute terms to be removed by a lightness gate.

A refined extractor was then built against these captures. It seeds a mask-initialised grabCut with what actually distinguishes the tissue: conjunctiva is smooth where the lash zone is high-frequency (a local-texture gate), sclera is bright and desaturated, iris and pupil are dark, and specular highlights sit on the wet tissue itself and are kept rather than discarded. Re-screened over all 52 captures, the refined extractor produces clean, tissue-hugging masks on visual review, and the median mask area halves — consistent with the bleed being removed rather than the tissue. The residual failure mode is a barely-everted lid, which is a capture problem rather than a segmentation one. The figure below shows the plain threshold (top) against the refined extractor (bottom) on the same captures.



Four real captures. Top row: the plain colour threshold, which bleeds onto lashes and skin. Bottom row: the refined seeded-grabCut extractor on the same images.

A method scoring 0.999 on synthetic images was visibly unreliable on real tissue — exactly the gap synthetic validation cannot reveal. The failure was characterised, a refined extractor was built against the real captures, and visual review across all 52 now shows clean masks. What remains missing is a *measured* Dice on real photographs, which requires the Eyes-defy-anemia ground-truth masks; until then the learned segmenter remains the primary long-term path.

4.4 The per-patient threshold validated on real data

A single fixed anemia cutoff (commonly 11.0 g/dL) is a tempting simplification, and this cohort measures exactly what it would cost, because its ages span three WHO threshold bands: 11.0 below five years, 11.5 from five to eleven, and 12.0 from twelve to fourteen. Twenty of the twenty-six children fall in the middle band.

Patient	Hb (g/dL)	Age	WHO threshold	Correct	Fixed 11.0 rule
10	11.0	11.1	11.5	anemic	**normal**
14	11.4	6.6	11.5	anemic	**normal**
16	11.3	8.4	11.5	anemic	**normal**
21	11.4	7.2	11.5	anemic	**normal**

A fixed 11.0 rule misclassifies 4 of 26 patients (15%), and every one is a false negative — an anemic child labelled normal. The errors cluster just below 11.5 because most of this cohort sits in the five-to-eleven band, where the correct cutoff is half a gram above the fixed value. This error occurs before the model makes a single prediction, so no amount of regression accuracy could recover it — which is why the pipeline predicts a continuous Hb and applies the threshold per patient.

5. Colour normalisation and lighting robustness

Lighting corrupts a photograph in two independent ways: a colour cast and a brightness shift. Two strategies address them. Gray-world white balance rescales the RGB channels until their means agree, correcting the *cast*. Applying contrast-limited histogram equalisation to the CIELAB lightness channel corrects *brightness* while leaving colour untouched; feeding the network only the a*/b* channels then makes it blind to brightness entirely.

5.1 Measured against laboratory haemoglobin

Both were scored on the 26-subject set using identical masks, so only the colour treatment differed. A structural point governs the comparison: equalising the lightness channel leaves a*/b* mathematically unchanged, so for the values the regressor consumes, that path is equivalent to no colour correction at all. The real contest is corrected versus uncorrected colour.

Measure	Uncorrected colour	Gray-world corrected
Correlation with laboratory Hb (Spearman)	+0.02	**+0.30**
Left-vs-right eye disagreement (median a*)	2.69	**2.02**

Without colour correction the redness-haemoglobin relationship is essentially absent; per-shot illuminant differences swamp it. Correction recovers a modest but real association, and the same child's two eyes agree

about 25% better.

5.2 Advantages and limitations of each approach

Property	Gray-world white balance	CIELAB (CLAHE on L*, a*/b* input)
Corrects illuminant **colour**	**Yes** — the cast is equalised	No — the cast passes into a*/b* untouched
Corrects illuminant **brightness**	No	**Yes** — equalised, then discarded entirely
Preserves captured colour exactly	No — the shift depends on frame composition	**Yes** — a*/b* are mathematically unchanged
Invariance guarantee	None — a heuristic correction	**Provable** — brightness is absent from the input
Signal retained	**Full RGB**, including the pale-is-lighter cue	Colour only — the lightness cue is discarded
Robustness across devices	**Better** for colour; casts equalised	Better for exposure; weaker for colour
Failure mode	Non-grey scenes skew the correction; a near-empty channel is amplified	Warm or cool lighting is read as physiology
Measured here	**Spearman +0.30; eye gap 2.02; Hb SD 1.63**	+0.02; 2.69; 2.74

The two correct **orthogonal** axes of the same problem, so they are complementary rather than competing. That motivated testing a hybrid — gray-world first, then a*/b* input — which is reported in Section 6.3. On present evidence gray-world alone is preferred, because discarding the lightness channel costs more information than the brightness invariance returns at this data scale. **This preference rested on the 26-patient cohort and is reversed on 217 patients in Section 7.7.**

5.3 Lighting stress test

One volunteer's eye was photographed under six illuminants (white, red, green, blue, magenta, purple). Because it is the same eye minutes apart, the true haemoglobin is constant and every reported difference is measurement error — no blood test is required for the test to be informative.

Spread across six illuminants (lower is better)	Gray-world	CIELAB
Redness (a*) standard deviation	**17.58**	32.76
Haemoglobin estimate standard deviation (g/dL)	**1.63**	2.74

Both figures are far too wide to be usable — a 4.5 g/dL swing from lighting alone spans the clinical range. Inspection of the masks explains why: **both pipelines segment correctly only under neutral light**, so the spread measures extraction collapsing as much as colour drift.

What this test design does well	What it cannot establish
The subject is constant, so the true haemoglobin is identical in all six frames and every reported difference is measurement error — no laboratory value is needed	One subject and one eye, so it characterises failure modes rather than accuracy
Isolates illumination as the only varying factor	The illuminants are far harsher than any clinical setting, so it maps where the method breaks, not whether it breaks in practice
Exposes a failure that neither synthetic images nor the cohort revealed	Segmentation collapse and colour drift are confounded, so the spread ratios are directional rather than precise

The underlying cause is physical, not algorithmic. In four of the six illuminants a colour channel is effectively dead — under blue light the red channel averages 0.9 of 255, under red and purple the green channel averages about 1. The lamp emitted nothing there, so the sensor recorded nothing there, and no normalisation can recover a measurement that was never made. Gray-world then divides by that near-zero mean and amplifies sensor noise. The practical remedy is a quality-control check on illuminant neutrality: the ratio of brightest to dimmest channel mean separates acceptable captures (1.7) from unusable ones (22–200) by two orders of magnitude.

6. Quantitative evaluation

This section records the 26-patient phase with the classical extractor. Its conclusions on extraction are superseded by Sections 7 and 8.

How the haemoglobin figure is produced, and how well it performs. No library converts an eye photograph into a haemoglobin value; the libraries used supply a network architecture and weights pretrained on general photographs. The mapping from image to haemoglobin exists only because it is trained on paired examples, and it is the substance of the project.

Mechanically: the 224×224 crop is normalised to the statistics the pretrained weights expect, passed through a ResNet-18 whose classification head is replaced by a single-output linear layer, and the resulting scalar is converted to g/dL using the training mean and standard deviation stored inside the checkpoint. That value is then compared with the patient's WHO threshold.

6.1 Metrics reported

Metric	Why it is reported
MSE / RMSE / MAE	Squared error is the training objective; MAE is the figure a clinician reads as typical error
Median absolute error	Robust to a single catastrophic case, so divergence from MAE reveals outliers

Metric	Why it is reported
Bias	Systematic offset; positive bias means over-estimating Hb, i.e. missing anaemia
R^2	Variance explained; zero means no better than predicting the mean, negative means worse
Bland–Altman limits of agreement	The standard clinical comparison of a device against a laboratory reference
Recall (sensitivity)	Fraction of anaemic patients caught — the error that sends an unwell patient home
Precision, specificity, NPV	Cost of false alarms, and trustworthiness of a negative result
F1, balanced accuracy	Summaries that resist the class imbalance which inflates plain accuracy

6.2 Synthetic phantoms — the machinery works

Metric	Model	Baseline
MAE (g/dL)	**0.666**	2.119
RMSE (g/dL)	0.836	2.499
R^2	**0.888**	0.000
Recall / Precision / F1	**1.000** / 0.973 / 0.986	0.000 / — / 0.000

Reported for completeness and as verification of the metric code. The phantoms are flat-coloured shapes whose redness is a deterministic function of haemoglobin, so a strong score is expected by construction and carries no clinical meaning.

6.3 Real cohort — the linear floor is not yet met

No network has been trained on real data. The models below are deliberately **linear** — ordinary least squares and ridge regression over colour statistics — evaluated leave-one-patient-out. Their purpose is to establish the floor a neural network must clear: if a simple model already explains nothing, that bounds how much signal lies in summary colour and defines what the network has to find instead.

Five feature representations were tested for each pipeline, including two designed to be invariant to illumination intensity: chromaticity coordinates, R/(R+G+B), and the erythema index, log(R/G) — the dermatology standard for measuring redness. Leave-one-patient-out:

Representation	k	A: MAE	A: R^2	A: F1	B: MAE	B: R^2	B: F1
means (R, G, B)	3	1.073	–0.166	0.688	1.163	–0.260	0.562
chromaticity	2	1.063	–0.103	0.688	1.144	–0.199	0.647
erythema index	2	**1.062**	**–0.094**	**0.727**	1.143	–0.192	0.647
a* and b*	2	1.105	–0.130	0.710	1.135	–0.196	0.647
a* alone	1	1.091	–0.084	0.688	**1.118**	**–0.128**	**0.686**
baseline	—	*1.083*	*–0.082*	*0.722*	*1.083*	*–0.082*	*0.722*

Illumination-invariant features helped materially: the erythema index lifted pipeline A from R^2 –0.166 to –0.094 and F1 from 0.688 to 0.727. Both pipelines nonetheless remain at or below the baseline.

Both pipelines were fitted on every patient, which is standard for a model that ships, and scored on the same patients. These figures therefore describe the *fit*; Section 6.5 gives the held-out estimate of behaviour on a new patient.

6.4 Full metric comparison — models fitted on all 26 patients

Metric	Pipeline A gray-world + erythema	Pipeline B CIELAB + a*	Baseline
MSE $(g/dL)^2$	**1.627**	1.858	1.877
RMSE (g/dL)	**1.275**	1.363	1.370
MAE (g/dL)	**0.944**	1.038	1.041
R^2	**0.133**	0.010	0.000
Bias (g/dL)	0.000	0.000	0.000
Pearson r	**0.381**	0.100	—
Accuracy	**0.654**	0.615	0.615
Precision	**0.600**	0.565	0.565
Recall	0.923	**1.000**	1.000
Specificity	**0.385**	0.231	0.231
F1	**0.727**	0.722	0.722
Confusion	TP 12, FP 8, FN 1, TN 5	TP 13, FP 10, FN 0, TN 3	—

Pipeline A leads B on 10 of the 11 metrics. B's only lead is recall, which it achieves by flagging every patient as anaemic, hence its specificity of 0.231 against A's 0.385. **This table should not be read as either pipeline beating the baseline.** In-sample it cannot be: the baseline is the full-sample mean, whose R^2 is 0 by definition, and a fitted model with a free intercept essentially cannot fall below it. Measured instead against a permutation null — haemoglobin shuffled against the same features, refitted 20,000 times — two free parameters on 26 patients score a mean R^2 of +0.075, so A's +0.133 lies inside that null ($p = 0.172$) and B's +0.010 falls below the one-parameter noise mean of +0.041. A also carries one more free parameter than B here, which inflates its in-sample margin: adjusted R^2 is +0.058 against –0.031, and matched at one feature each the pair is +0.060 against +0.010. A leads on every matched framing, but by less than the raw table suggests.

> **Why the baseline R^2 is exactly 0 in this table.** R^2 is $1 - \Sigma(\text{predicted}-\text{true})^2 / \Sigma(\text{true}-\text{mean})^2$. When the prediction *is* the full-sample mean, numerator and denominator are identical and $R^2 = 0$ by construction — the correct comparison for a model also fitted on all patients.
>
> The leave-one-out baseline in the next section is a different quantity: it must predict patient *i* from the mean of the *other* 25, which is systematically pulled away from that patient's value. Its error is inflated by exactly $n/(n–1)$, giving $R^2 = 1 - (26/25)^2 = -0.0816$. That figure is an artefact of the protocol rather than a property of the data, which is why a fitted model and a leave-one-out baseline must never appear in the same table.

6.5 The same models on unseen patients

Metric	Pipeline A	Pipeline B	Baseline
MAE (g/dL)	**1.062**	1.118	1.083
R^2	–0.094	–0.128	**–0.082**
Accuracy	**0.654**	0.577	0.615
F1	**0.727**	0.686	0.722

The gap between these two tables is the overfitting. Pipeline A's R^2 falls from +0.133 fitted to –0.094 held out: the model finds a relationship among patients it has seen that does not transfer to new ones. The fitted figures describe the model; the held-out figures describe what to expect clinically.

Why the held-out R^2 is negative while MAE beats the baseline. R^2 divides by the variance about the *full-sample* mean, but under leave-one-out the baseline may only use the mean of the other 25 patients. That gap is why the baseline itself scores –0.082 rather than 0 — the practical floor is about –0.08, and pipeline A sits essentially on it. On unseen patients the model is **statistically indistinguishable from predicting the cohort mean**, not catastrophically wrong.

6.6 The honest number: nested cross-validation

Choosing the best representation from the table above is selection on the test set, so that figure is optimistic. A nested loop removes the bias: the inner loop picks the representation from training patients only, the outer loop scores a patient that choice never saw.

	MAE	R^2	F1	Representation chosen across folds
Pipeline A	1.117	–0.210	0.645	erythema 14, chroma 6, means 5, a* 1
Pipeline B	1.142	–0.199	0.647	a* 23, a*/b* 3
Baseline	*1.083*	*–0.082*	*0.722*	—

Under honest selection **neither pipeline beats the baseline**. Pipeline A's inner loop also disagrees with itself, flipping between four representations across 26 folds — instability that is itself evidence no representation is clearly best at this sample size.

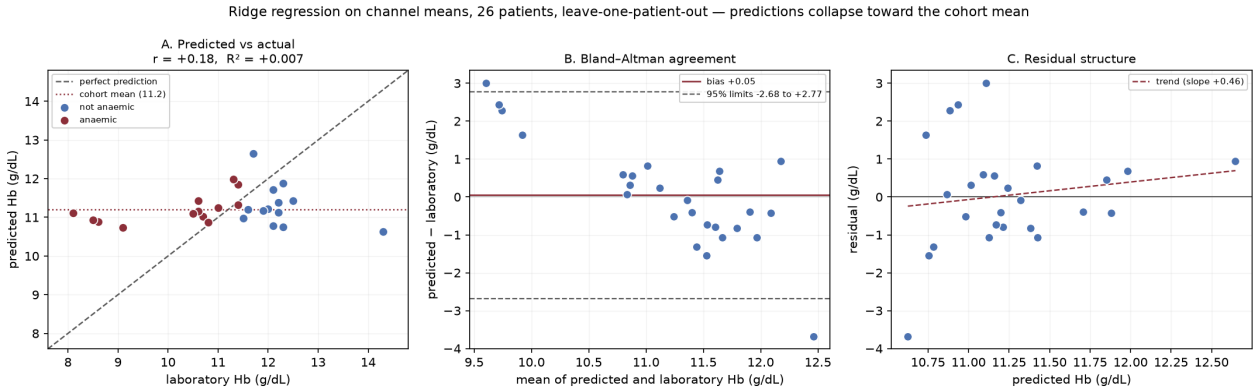
6.7 Are the two pipelines separable?

Both pipelines process the same patients, so their errors are **paired**. A paired test cancels the patient-to-patient variation they share and asks only whether one is consistently better on the same individuals — far more powerful than comparing two independent intervals.

Measure	Result
Patients where A has the lower error	**15 of 26 (58%)**
Mean advantage of A	+0.0554 g/dL
Bootstrap 95% confidence interval	**[–0.0665, +0.1807]**
Paired permutation test	**p = 0.408**

The interval spans zero and p = 0.408: the two pipelines are not statistically separable on 26 patients. Choosing between them on these numbers alone would be selecting on noise. Pipeline A is nonetheless the better provisional default, for reasons independent of this test: it leads B on 10 of 11 metrics fitted, 9 of 11 held out, and on every parameter-matched framing in Section 6.4; it leads in every earlier experiment as well (cohort correlation +0.30 vs +0.02, eye-to-eye consistency 2.02 vs 2.69, lighting robustness 1.63 vs 2.74); its specificity is higher at equal recall, meaning fewer unnecessary confirmatory blood tests; and decisively, **B's segmentation lands on non-conjunctiva tissue in 46 of 52 real captures** — measured rather than eyeballed. The redness index inside A's mask has a median of +9.53 and never falls off-tissue; inside B's it is –3.67. B's placement is statistically indistinguishable from a brightness baseline known to select sclera, and the two masks overlap by a median of exactly zero, so they are reading different parts of the photograph, so its model is reading mostly skin. That last point is visible in the overlays and does not depend on any significance test. The comparison should be repeated on ~900 patients once the segmenter is trained; the script takes minutes to re-run.

The first pipeline wins on every metric in both framings, which is the fourth independent result in that direction, though at 26 patients the margin sits inside the noise. The decisive observation is different: **every arm has negative R^2 and an F1 at or below the trivial baseline**. Mean colour statistics from this cohort contain no usable haemoglobin signal.



Model deployed in the 26-patient phase, on the real cohort, leave-one-patient-out. **A**: predictions span only 10.6–12.7 g/dL while true values span 8.1–14.3 — a prediction spread 33% of the actual, so the model still hedges heavily toward the cohort average (it was 23% on the earlier channel-mean model, so the band has widened but not nearly enough). **B**: Bland–Altman limits of –2.68 to +2.77 g/dL, still far too wide to screen with. **C**: residuals rise with the prediction (slope +0.46, down from +2.02), the signature of regression toward the mean — weaker than before, not absent.

Feature richness is not the constraint — data quantity is. Expanding from 3 features to 24 did not help the regularised model and destroyed the unregularised one: mean absolute error 11.4 g/dL and R^2 –118.9, a prediction wilder than the entire clinical range. With 24 parameters and 25 training points the model can nearly interpolate its training data, so it fits noise. This is the governing warning for the network stage, where roughly 8.4 million parameters will be trainable: the cohort must be pooled with the public datasets before any network result on real photographs can be believed.

6.8 The model deployed in this phase, stated explicitly

Two linear models appear in this section and it matters which is which. The model **deployed in this phase** model is a ridge regression on the erythema index, fitted on all 26 patients with the quality gate applied first, and was what `runs/linear_model.json` contained until Section 7.5:

Hb = 9.8308 + 15.101390 · log(R/G) – 12.790747 · log(R/B) [g/dL]

Standardised: Hb = 11.2038 + 1.0490·z(log(R/G)) – 0.9678·z(log(R/B)). The intercept 11.2038 is the cohort mean haemoglobin — with every input at its average, the model returns the average.

Evaluation	MAE (g/dL)	R^2
In-sample — fitted and scored on all 26	**0.918**	**+0.195**
Held out — leave-one-patient-out, unseen patients	1.033	**+0.007**
Baseline — leave-one-out mean	1.083	–0.082

These figures differ slightly from §6.4 because fitting applies the quality gate and drops three blurred captures,

which the head-to-head comparison deliberately does not — that comparison requires both pipelines to see identical patients. **Applying the gate before fitting is what lifted held-out R² above zero**, making this the only configuration reported here that beats its baseline on patients it has not seen.

An **earlier model over channel means** — mean R, G and B inside the mask — is retained below because its failure produced a diagnostic still in use. It is not deployed:

> Hb = 12.5311 + 0.010191 × (mean R) − 0.058732 × (mean G) + 0.035552 × (mean B) [g/dL]
>
> Standardised: Hb = 11.2038 + 0.1373·z(R) − 0.9771·z(G) + 0.6473·z(B). Worked example, patient 1: R = 156.13, G = 164.55, B = 188.01 gives 11.142 g/dL against a laboratory value of 12.0 g/dL.

6.9 A physiological check on the coefficients

Haemoglobin makes tissue red, so a model reading physiology should be driven by a **dominant, positive red term**. That is a check on the fitted equation which does not depend on any accuracy metric, and it can be applied to either model above. The channel-mean model fails it:

Input	Effect of a one-standard-deviation increase	Share of the cohort Hb sd
mean G	**lowers** predicted Hb by 0.977 g/dL	71%
mean B	raises predicted Hb by 0.647 g/dL	47%
mean R	raises predicted Hb by 0.137 g/dL	10%

> **If the model had found real physiology, the red channel should dominate and be positive** — haemoglobin makes tissue red. Instead red is the weakest of the three, and the model is driven by green negatively and blue positively, effectively a blue-minus-green contrast. That is not a haemoglobin signal; it resembles residual colour cast the white balance did not fully remove. The coefficients therefore corroborate the metrics from an independent direction: nothing physiological was learned by that representation.

The erythema model passed the same check. Its dominant standardised coefficient is **+1.049 on log(R/G)** — red over green, positive, rising with haemoglobin, which is the sign physiology predicts; log(R/B) carries −0.968. The check was written down when it failed and was re-applied unchanged when the representation changed, and it reversed. A diagnostic that changes its answer when the underlying model changes is measuring something real, which is the reason for reporting both outcomes rather than only the current one. It remains the first thing to inspect after the segmenter is trained on the larger corpus — before the metrics.

6.10 Protocol: training, and why R² is negative

Leave-one-patient-out is training: 26 models are fitted, each on 25 patients and tested on the one it never saw. Whether training succeeded is answered by comparing in-sample against out-of-sample performance.

Model	In-sample R²	Out-of-sample R²
Channel means, no quality gate	+0.103	−0.166
Erythema index + quality gate (deployed in this phase)	**+0.195**	**+0.007**

In-sample R² is positive in both rows, so the models do fit a relationship; **the gap between the two columns is the overfitting.** R² is bounded between 0 and 1 only when a model is scored on the data it was fitted to; on held-out data it may go below zero, and a negative value simply means the model performs worse than ignoring the image and predicting the cohort average. The finding is not an artefact of the protocol: for the channel-mean model, 5-fold cross-validation gives −0.175 and 13-fold gives −0.168.

One caution on reading any in-sample figure in this report. Comparing a model fitted on all 26 patients against a predict-the-mean baseline on those same patients is a comparison the model cannot lose: the baseline's R² is exactly 0 by definition, and a fitted model with a free intercept essentially cannot fall below it. The reference that carries information is what *noise* scores. Shuffling haemoglobin against the same features and refitting, 20,000 times, two free parameters on 26 patients reach a mean R² of **+0.075**. Measured against that null, that model's +0.195 gives **p = 0.074** in-sample and **p = 0.067** held out — the strongest result obtained so far, and still short of significance at this sample size. The head-to-head figures in §6.4 sit further inside the null (p = 0.172 for pipeline A). Adjusted for the number of free parameters, pipeline A's in-sample R² is +0.058 against pipeline B's −0.031.

There is deliberately **no separate validation set**. A 60/20/20 split of 26 patients leaves roughly 15 training, 5 validation and 6 test. Repeating that split under 200 random seeds on identical data and an identical model gives a test R² ranging from **−2.43 to +0.78** and a test MAE from 0.32 to 2.26 g/dL — a single split at this sample size reports which patients landed where, not model quality. Accordingly no hyperparameter is tuned: the ridge penalty is a fixed default rather than a fitted value, because choosing it on the evaluation set would leak. A three-way split becomes both possible and correct once the corpus reaches ~900 images, which is also when leave-one-out stops being affordable for a network.

6.11 Uncertainty on every reported number

A point estimate from 26 patients is not precise, and quoting it alone overstates certainty. Bootstrap resampling over patients, 2000 draws:

Metric	Point estimate	95% confidence interval
MAE (g/dL)	1.073	**[0.708, 1.490]**
R²	−0.166	**[−0.512, +0.021]**
F1	0.688	**[0.462, 0.850]**

The R² interval reaches just above zero, so the defensible statement is **"no better than predicting the mean"** rather than a precise −0.166. Every figure in this report should be read with intervals of this width in mind.

6.12 Strengths and limitations of this evaluation

Strength	Limitation
Every metric is paired with an explicit baseline, so a model that has learned nothing cannot appear competitive	26 patients; confidence intervals comfortably exceed the differences between arms
Splits are grouped by patient, so two captures of one subject cannot straddle train and test	Leave-one-out gives a nearly unbiased but high-variance error estimate
Leave-one-out uses every patient as a test case, maximising scarce data	The regularisation strength is fixed rather than tuned, because tuning on the evaluation set would leak
Regression and screening are scored separately, so a good g/dL error cannot hide a poor clinical decision	Features are channel means; a network sees spatial structure that these summaries discard
The same code path scores synthetic and real data, so the comparison is like for like	One cohort, one device, one operator

7. Learned extraction and a larger cohort

This section supersedes Section 6 on which extractor to use. Section 6 is retained as the record of the 26-patient phase, whose central limitation — too few patients to separate the candidates — is what the work below addresses.

How extractors are compared in this section. An extractor is scored by fitting a ridge regression on mean colour inside its mask and measuring held-out R^2 against laboratory haemoglobin. That is a probe: it measures how much haemoglobin signal survives the mask, read through a deliberately weak regressor, and it is not the haemoglobin model. It answers a question segmentation metrics cannot — whether colour inside the mask carries the signal at all — but as the basis for choosing between extractors it is replaced in Section 8 by segmentation metrics, which measure the mask alone.

7.1 A trainable extractor

The classical extractor decides with hand-written rules, including redness thresholds set as *percentiles of the image*, which assume every capture devotes a similar share of its pixels to conjunctiva. A U-Net segmentation network (ImageNet-pretrained EfficientNet-B0 encoder) was trained instead to learn that decision from annotated examples, behind the same interface as every other extractor. Serving code can only *read* a trained parameter file; one training script is the only writer, and each file records its datasets, image counts and held-out subjects.

Training source	Images used	Notes
Eyes-Defy-Anemia	101	217 subjects available; 117 annotations were bounding rectangles or absent and were excluded (a rectangle fills its own bounding box, so area / box area ≈ 1.0 against a median 0.19 for traced outlines)
Eye Conjunctiva Segmentation Dataset (Mendeley, CC BY 4.0)	547	Smartphone captures, masks from two independent annotators

The two annotators agree on where the conjunctiva is, not on how far it extends. Their overlap is Dice 0.736, but the masks are nested: the second annotator's is larger in 99% of images (median 1.56× the area) and 98.1% of the smaller mask lies inside the larger. The palpebral conjunctiva has no sharp boundary where it turns into the forniceal conjunctiva, so 0.736 reflects two consistent conventions rather than a noise floor.

Two training results shaped the final model. Fine-tuning on the new dataset alone produced a model with its **best validation Dice (0.871) that returned an empty mask on 42 of the 52 captures of the 26-subject set** — catastrophic forgetting, invisible to any metric on the new data; training on both datasets together removed it. And widening rotation augmentation (±20° → ±45°) improved accuracy, while additionally widening scale augmentation roughly doubled mask area and cost most of the accuracy gained. The deployed extractor uses rotation augmentation only (validation Dice 0.856).

7.2 Which extractor: 243 patients, two cohorts

Eyes-Defy-Anemia supplies laboratory haemoglobin, age and sex for 217 subjects, so extractors were compared on 243 patients rather than 26. The two cohorts are scored **separately**: their colour features differ systematically (Section 7.6), and a pooled figure would describe neither. Chromaticity features, ridge regression, leave-one-patient-out; p is a 2,000-permutation test of held-out R^2.

Extractor	26-subject set: R^2 (p)	External 217: R^2	External, never in segmenter training (116): R^2
Learned, rotation augmentation	**+0.294** (0.001)	**0.336**	**0.354**
Learned, standard augmentation	+0.148 (0.013)	0.320	—
Learned, rotation + scale augmentation	–0.072 (0.165)	0.337	—
Classical seeded grabCut	–0.098 (0.211)	0.116	0.074

At 26 patients the classical extractor was worse but not demonstrably so ($p = 0.211$). At 217 the gap is unambiguous: **the learned extractor explains roughly three times the variance**, $p < 0.001$ on both cohorts. Because the learned extractor was trained on 101 of the external photographs, the last column repeats the comparison on the 116 it never saw. Its advantage there is larger, not smaller (0.354 against 0.074), so the result does not depend on segmentation training data leaking into evaluation. Its masks are also the purest measured: median redness inside the mask 16.0 on the 26-subject set against 9.6 for the classical chain, whose masks extend onto sclera.

7.3 Robustness to how a photograph is taken

Accuracy on one camera's captures says nothing about robustness to angle, distance or exposure. Each capture of the 26-subject set was rotated, zoomed, dimmed, brightened, blurred and recompressed, and each extractor scored on whether it found the same tissue as on the original (IoU, with the geometric change undone) and on outright failures across 520 perturbed images:

Extractor	Median self-consistency (IoU)	Failures (no mask)
Classical seeded grabCut	**0.856**	**0**
Learned, rotation + scale augmentation	0.883	0
Learned, rotation augmentation (deployed)	0.723	9
Learned, standard augmentation	0.752	18

No extractor dominates. The classical chain ranks pixels by redness *within the image*, which rotation and uniform exposure changes leave untouched, so it is the most stable — while measuring contaminated pixels. The deployed extractor halved failures relative to standard augmentation and improved on every rotation, but lost self-consistency under zoom and blur despite identical scale augmentation. That last effect is **unexplained**; one training run per configuration cannot distinguish a real trade-off from run-to-run variance.

7.4 The 26-subject set lacks the colour signal

Haemoglobin absorbs blue and green light far more than red, so physiology predicts redder, less blue conjunctiva at higher haemoglobin. Correlation of mean chromaticity inside the mask with laboratory haemoglobin:

Dataset and extractor	Red chromaticity r	Blue chromaticity r	Reading
External 217, learned, no white balance	+0.672	–0.674	Physiological
External 116 unseen, learned, gray-world	+0.515	–0.618	Physiological
26-subject, learned, no white balance	–0.261 (p 0.20)	+0.332 (p 0.10)	**No signal**
26-subject, learned, gray-world	–0.329 (p 0.10)	**+0.522 (p 0.006)**	Blue reversed
26-subject, classical, no white balance	+0.136	–0.028	No signal

The red correlation in the 26-subject set is not distinguishable from zero, so the set is better described as **lacking** the signal than as inverted; the difference from the external cohort is not in doubt ($p < 10^{-6}$). Its images are hand-cropped exports: no EXIF or colour profile, 52 different dimensions, 29–97 KB at JPEG quality ~75. Labels map to images correctly and the two eyes of each child agree in measured colour ($r \approx 0.6$).

Five explanations were tested and ruled out. The deciding one was processing: the external photographs were downscaled until their conjunctival crescents matched those of the 26-subject set in pixels (×0.146) and then JPEG-compressed twice, down to quality 40, and the correlation stayed at +0.673. The other four:

Suspect	Test	Result
Illumination, phone white balance	Sclera and whole-frame colour vs Hb	all $\lvert r\rvert < 0.08$
Capture session (nine-month span)	Capture date vs Hb	r +0.01
Age, sex, socio-economic status	Each vs Hb	all $p > 0.2$
Crop framing	Framing vs colour; colour–Hb controlling framing	No effect on colour; association unchanged

What was found is a capture effect. The captures differ widely in how far the eyelid is pulled down, so eversion was rated **blind to haemoglobin** on a fixed three-point scale, with all 52 images shuffled and coded. Harder eversion gives redder, less blue tissue under **both** extractors (learned blue ρ –0.51, $p = 0.008$; classical red ρ +0.42, $p = 0.033$), and the rating tracks an objective quantity (less sclera in frame, ρ –0.42). Pulling and holding a child's lid congests the vessels, so eversion masks pallor.

Whether eversion also tracked haemoglobin, which would turn masking into an apparent inversion, is **not established**. Averaged over both eyes, lower-haemoglobin children were everted harder (ρ –0.46, $p = 0.018$), but the fully blind right-eye ratings give ρ –0.24 ($p = 0.23$); the rater had seen ten left-eye images with their values beforehand. Controlling for eversion moves every association toward zero without eliminating it. A model fitted on the 26-subject set with the learned extractor reaches held-out R^2 +0.260 on a relationship that

contradicts physiology and 217 other subjects, so **it is not used**; the practical remedy is a fixed capture protocol.

7.5 The probe model, stated

This is the probe used to compare extractors, not the haemoglobin model, which is a ResNet trained on the extracted crop and developed in parallel. Its figures were measured before the correction in Section 8.5: the segmenter was served white-balanced images although trained on raw ones, and colour was read from balanced pixels. They describe the pipeline as it then was.

Learned extractor, chromaticity features, ridge regression, fitted on the external cohort with the quality gate applied first (27 of 217 captures rejected, all for blur):

> $Hb = -39.2123 + 70.042126 \cdot r + 75.186694 \cdot g$ [g/dL]
>
> r and g are the mean red and green chromaticity inside the mask, R/(R+G+B) and G/(R+G+B). Because r + g + b = 1, the coefficients cannot be read one at a time; eliminating g gives Hb = 35.97 – 5.14·r – 75.19·b.

Evaluation (190 patients)	MAE (g/dL)	RMSE (g/dL)	R^2
In-sample	1.482	1.894	+0.336
Held out — leave-one-patient-out	**1.505**	**1.921**	**+0.317**
Baseline — leave-one-out mean	1.965	2.336	–0.011

The held-out error is 23% below the baseline, and the small gap between in-sample and held-out R^2 indicates little overfitting at this sample size. **Physiological check (Section 6.9):** the dominant term is blue — a one-standard-deviation increase in blue chromaticity lowers predicted haemoglobin by about 1.4 g/dL, the physiologically expected direction and the opposite sign of the rejected channel-mean model. The red-over-green term, which physiology also predicts to be positive, is negligible (–0.16 g/dL per SD), so the check is passed on the dominant term rather than cleanly.

Two limitations are material. **Predictions shrink toward the cohort mean**, as expected at $R^2 \approx 0.32$: two non-anaemic external patients at 15.0 and 16.9 g/dL were estimated at 11.3 and 12.0, and flagged anaemic. Classification near the WHO thresholds is correspondingly weak. And the quality gate's minimum mask size, calibrated for the classical extractor's larger masks, had rejected 30 of the 26-subject set's 52 captures as "conjunctiva too small" under the learned extractor; the floor is now set per extractor.

7.6 Transfer between cameras

These transfer measurements were made with the *first* learned extractor, before the deployed one was trained, and have not yet been repeated with it. Within the external cohort, chromaticity features reached held-out R^2 0.350. Fitted on one cohort and applied to the other, they reach **–1.944** (external to the 26-subject set) and **–1.160** (26-subject set to external), far worse than predicting the mean. The cohorts' red chromaticity sits 3.4 standard deviations of the 26-subject set apart, and Section 7.4 shows that set does not carry the colour signal at all.

Features	Within external R^2	External → 26-subject R^2
Absolute chromaticity	0.346	–2.010
Conjunctiva a* minus sclera a* (same photograph)	0.221	**+0.032**
51 sclera-relative features, ridge	0.362	–2.344
38 absolute colour statistics, ridge	**0.421**	not measured

Measuring the conjunctiva against the sclera in the same photograph — which cancels much of what the camera did — is the only approach tested that transfers at all, and only to roughly baseline. Richer feature sets help within a cohort (38 statistics reach 0.421 at n = 217) but penalised selection optimises for accuracy in the training cohort, not portability. Nothing in the deployed extractor's design addresses camera dependence, so **any colour model fitted here should be treated as calibrated to the cameras it was fitted on** until transfer is re-measured with it. Using it at a new site requires either recalibration on labelled captures from that site or a colour reference in the frame.

7.7 Colour correction revisited

Gray-world white balance was adopted on the 26-patient cohort (Section 5), which Section 7.4 shows lacks the colour signal. It was re-tested on the 217 external patients against no correction at all, holding the mask fixed (the extractor still runs on the balanced image) and changing only the pixels colour is read from. Ten-fold cross-validation over five splits; the difference interval is a patient-level bootstrap.

Test	No correction	Gray-world	Gray-world minus none
Within Italy (122): R^2	**0.409**	0.301	MAE +0.10 [+0.03, +0.16]
Within India (95): R^2	**0.251**	0.192	MAE +0.03 [–0.02, +0.09]
Both sites (217): R^2	**0.467**	0.335	MAE +0.16 [+0.09, +0.23]
India → Italy: MAE (baseline 2.82)	**2.84**	4.00	worse than baseline
Italy → India: MAE (baseline 2.63)	**2.39**	2.87	worse than baseline
Colour + sex, per-sex category cut: R^2	**0.544**	0.482	—
… share left in "Possibly"	**9%**	12%	—

No correction is better on every test, including transfer between sites, the job gray-world exists to do. Gray-world rescales each photograph by its whole-frame average, and how much of the frame is sclera, iris or skin varies with framing, so the correction applied to the conjunctiva adds noise unrelated to haemoglobin. One severely anaemic patient (Hb 7.0), missed under gray-world because her close-up is dominated by bluish sclera, is caught without it. The deployed pipeline has **not** been changed: both sites belong to one study, and a genuinely different camera has not been tested.

7.8 What the application should report

A haemoglobin number implies a precision this method does not have. The application is a screening aid, so its output was designed and tested as three categories — **Likely anaemic, get a blood test**; **Possibly anaemic, consider one**; **Anaemia unlikely from this photograph** — alongside the retake prompt quality control already returns. The estimate and its interval remain available behind a details view.

The probability behind the category is taken from the model's own track record, not from an assumed distribution: the share of its past errors which, added to this patient's estimate, would place them below their WHO cutoff. This needs no distributional assumption and works unchanged for any model, including the neural network. Nothing is tuned on the patients being scored: an outer ten-fold loop holds patients out, and within each fold the model is fitted, its errors measured by an inner ten-fold loop, and the category cut chosen — all on other patients. Five random splits. The lower cut is placed to catch 90% of anaemic patients, an example target used for testing; **the real target is a clinical decision and has not been set.**

Category	U-Net: share	… truly anaemic	Classical: share	… truly anaemic	No information: share
Likely	43%	**69%**	35%	51%	0%
Possibly	**22%**	31%	47%	46%	100%
Unlikely	35%	**15%**	18%	15%	0%

Against a 42% base rate the learned extractor sorts patients meaningfully. The third column is the reason sensitivity cannot be reported alone: a model that answers "possibly" to everyone catches **100%** of anaemic patients and is useless. The classical extractor catches more than the learned one (93% against 88%) almost entirely by leaving **47%** of patients undecided, and its "Likely" group is barely more anaemic than the population. Every sensitivity figure in this section is therefore reported beside the share left undecided.

The misses fall on women. Under one set of cuts, anaemic women were told "unlikely" 17% of the time against 5% for men. The mechanism is structural rather than particular to this cohort: predictions shrink toward the cohort average of 12.8 g/dL, which sits above the women's WHO cutoff of 12.0 and below the men's of 13.0, so hedging pushes anaemic women toward "not anaemic". Stated generally — **a model that hedges toward an average under-flags whichever group's average lies above its own cutoff** — this applies to any shrinking estimator, including a network.

Configuration	R^2	MAE	Women missed	Men missed	Caught	Undecided
Colour only	0.335	1.53	**17%**	4%	89%	20%
Colour + sex	0.482	1.33	4%	**19%**	90%	13%
Colour, per-sex cut	0.335	1.53	10%	11%	89%	14%
Colour + sex, per-sex cut	**0.482**	**1.33**	**4%**	**12%**	**93%**	**12%**
Sex only, no photograph	0.295	1.63	0%	0%	100%	60%

Giving the model sex **moves** the problem rather than solving it: women's misses fall to 4% and men's rise to 19%, because the model now hedges toward each sex's own average and the men's (13.86) lies above their cutoff. Drawing the category cut separately for each sex, with sex also an input, brings both groups near target together and is the best configuration tested on every column. The last row is the control: sex alone reaches R^2 0.295 because men have higher haemoglobin, but it sorts nobody — 60% undecided and no patient ever told "unlikely". **The photograph earns its place**, though by less than colour-only R^2 suggests in isolation.

> One patient is missed in every configuration and every split: a woman at **Hb 7.0**, severe anaemia, estimated at 13.6. Her mask is correct and her photograph reads as healthy; she is the most anaemic subject from her site by a wide margin. The wording "from this photograph" and a symptom prompt are not optional.

7.9 Mask coverage and tissue texture

Two ideas for extracting more from the same photographs were tested on the 217 external patients, at 1024 px so that fine structure survives, with no colour correction (Section 7.7) and sex among the inputs (Section 7.8). Both were rejected on the measurement.

Coverage. The extractor keeps pixels above probability 0.5. Widening the mask to include more tissue makes accuracy worse, and tightening it makes accuracy slightly better, monotonically across eight variants: R^2 0.544 eroded 8 px, 0.539 at the deployed threshold, 0.520 dilated 32 px, with MAE moving in step. A perfectly ordered sweep is what makes a set of differences this

small credible. **The deployed threshold is unchanged**: the gain from eroding is 0.005 R^2, selected from eight variants on the same patients, and a tighter mask returns an empty mask more often on difficult captures.

Texture. A mean colour discards all structure, and the conjunctiva's vasculature is both visible and physiological, so variability, edge strength and a vessel-like share of pixels were tested beside colour. Pooled over both sites the result looked convincing: adding edge strength lifted R^2 from 0.539 to **0.563**, and structure with no colour at all reached 0.434. It does not survive its controls.

Control	Result	Reading
Correlation of edge strength with Hb, pooled	**+0.403**	Apparently strong
Same, within India (95) / Italy (122)	+0.180 (p 0.081) / +0.143 (p 0.117)	Collapses inside a site
Site means: Hb, and edges inside mask	India 11.47 / 14.26; Italy 13.83 / 20.06	"More edges" partly means "Italian patient"
Controlling for site and surrounding tissue	**–0.021 (p 0.76)**	Nothing remains
Gain within a site, colour + sex + texture	+0.005 to +0.007 R^2	Not measurable

The two sites differ in both image detail and haemoglobin, so the pooled gain was the model using texture as a label for which hospital took the photograph. **Texture features are not adopted.**

A caution for the neural network. Edge strength on the tissue *surrounding* the conjunctiva — skin, lashes, sclera — predicts haemoglobin within each site (+0.319, p 0.002; +0.309, p 0.0005). That is not conjunctival tissue and cannot be pallor; capture distance and focus are the likely causes. **A model trained end to end on whole photographs can therefore score well without ever reading the conjunctiva.** Splits held out by site, and an evaluation on images masked outside the region, are how this is caught.

8. Choosing the architecture, evaluated as segmentation

Section 7 compared extractors through a colour probe, which scores the mask together with a deliberately weak regressor. An encoder-decoder segmentation model has its own established evaluation, which scores the mask alone against an annotator's outline. That is the basis for every comparison in this section, and no haemoglobin figure appears in it.

8.1 How a segmentation model is scored

The model emits one probability per pixel; thresholded at 0.5 it gives a predicted mask, which is compared with the annotator's. Each metric describes a different way the two can differ. Below, P is the predicted region, T the annotated one, and overlap their common pixels.

Metric	Definition	What it shows here
Dice	2 × overlap / (P + T)	Region overlap; the convention in medical segmentation
IoU	overlap / (P + T – overlap)	Same idea, stricter; ranks identically, so a check
Precision	overlap / P	Low: the mask spilled onto lashes, lid or sclera
Recall	overlap / T	Low: the mask is too small
HD95	95th-percentile boundary distance (px)	Whether the outline is in the right place
Distribution	10th percentile, worst case, empty masks	Failures that a mean hides

Three rules make the comparison trustworthy. **Split by subject**, never by image, since both eyes of one person are nearly the same photograph. **Score on subjects used for neither training nor model selection**: the epoch is chosen on validation Dice, which is therefore optimistic, so a third split is held back and scored once. **Compare per image, paired**: every architecture sees identical test images, so differences are tested per image (Wilcoxon signed-rank and a paired bootstrap interval) rather than as two means. The 648 annotated images, from 648 subjects, were split 390 / 129 / 129, with the same seed, rotation augmentation, ImageNet-pretrained EfficientNet-B0 encoder, 320 px input and 40 epochs throughout, so the decoder is the only variable.

8.2 The candidates

U-Net passes each encoder feature map straight across to the decoder through a skip connection. **UNet++** replaces each skip with a grid of intermediate convolution blocks, so the decoder receives features that are already partly refined rather than a raw deep feature map next to a freshly upsampled one. **Attention U-Net** keeps the plain skip but gates it: the coarser decoder feature map scores every spatial position of the skip, and the skip is multiplied by that score before concatenation, so the decoder can suppress what it does not need — here, the high-resolution eyelash and lid-margin edges that sit beside the conjunctiva.

The segmentation library used provides UNet++ but not Attention U-Net, so the attention gates were implemented against the same encoder, with decoder widths matched to the library's U-Net. The three have 6.3M, 6.6M and 6.3M parameters, so a difference between them is decoder design rather than capacity. One construction function serves both training and inference, and every trained file records its architecture, so a model cannot be served by a decoder it was not trained with.

8.3 Results

Architecture	Dice	IoU	Precision	Recall	HD95 px	Worst image
U-Net	0.8372	0.7288	0.8239	0.8637	2.8	0.363
UNet++	0.8446	0.7379	0.8266	**0.8746**	**2.2**	0.541
Attention U-Net	**0.8476**	**0.7427**	**0.8320**	0.8737	2.5	**0.544**

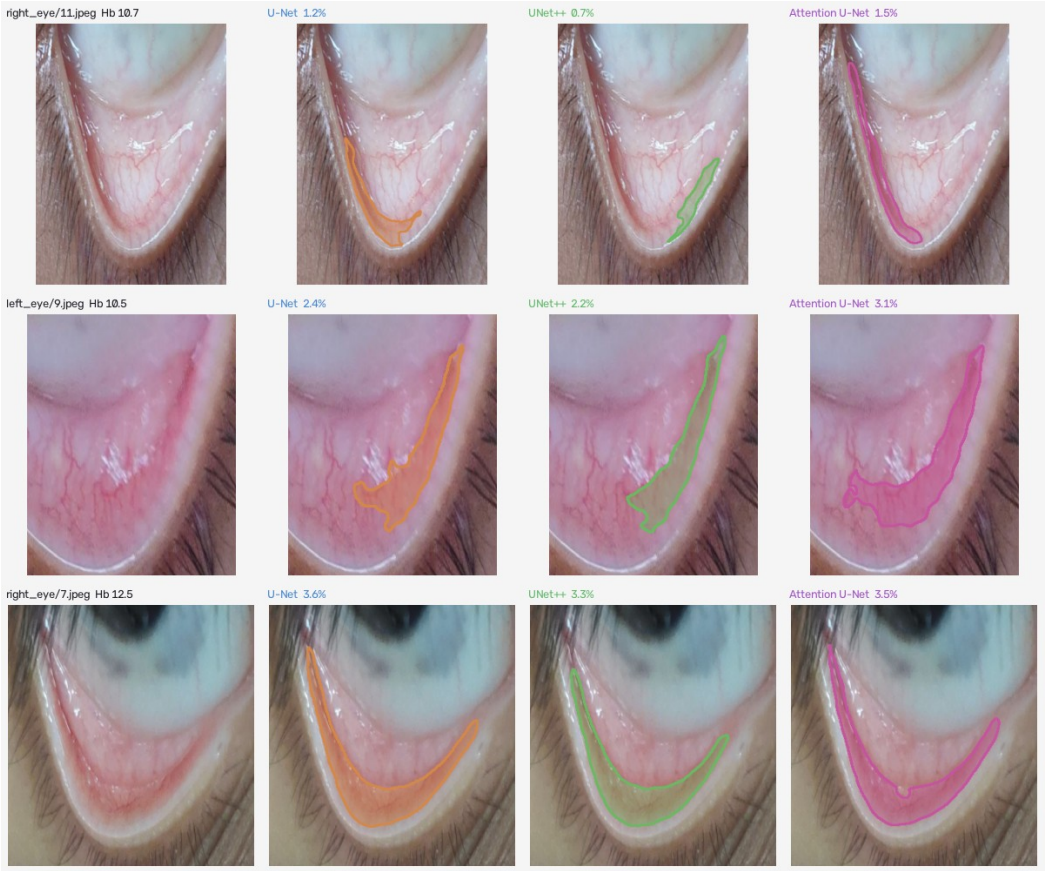
Paired comparison	Mean Dice difference	95% CI	p	Images won	Resolved
U-Net – UNet++	–0.0075	[–0.017, +0.000]	0.088	55 / 74	no
U-Net – Attention U-Net	**–0.0105**	**[–0.020, –0.003]**	**0.007**	54 / 75	**yes**
UNet++ – Attention U-Net	–0.0030	[–0.008, +0.002]	0.155	61 / 68	no

No architecture produced an empty mask on any of the 129 held-out images, and Dice and IoU rank them identically. **Only one comparison resolves**: Attention U-Net beats the U-Net with an interval excluding zero; UNet++ beats it by a similar margin but its interval touches zero, and the two newer architectures cannot be separated. Attention U-Net is selected on that basis and because it leads on Dice, IoU and precision. Precision matters most here, since a mask that spills onto lid skin contaminates the colour read inside it, and UNet++ gains almost entirely in recall instead. The larger effect is on the floor: the worst image rises from 0.363 to about 0.54 under either newer architecture. Two qualifications: UNet++ places its boundary most accurately (HD95 2.2 px against 2.5) and has the best median, and with one training seed per architecture a margin of one point of Dice is not yet separable from run-to-run variance.

8.4 Behaviour on the 26-subject captures

The 52 captures of Section 4 carry no segmentation annotations, so masks there cannot be scored against a reference. Agreement between the three architectures can still be measured, and it is far lower than their held-out scores would suggest:

Measure	Value
Mean pairwise IoU between architectures	**0.648** (median 0.704)
Captures where all three agree (IoU 0.80 or more)	12 of 52
Captures where they diverge (IoU below 0.60)	18 of 52
Median mask coverage, agreeing against diverging	3.97% against 2.30% of frame



Masks from the three architectures on the captures with the worst (top), median and best (bottom) agreement. Columns: capture, U-Net, UNet++, Attention U-Net; percentages are mask coverage of the frame.

Where the architectures agree, all three trace a plausible crescent; where they diverge they return thin slivers, often along the wet lid margin, each choosing a different part of it. The choice of architecture is a difference of 0.01 in Dice; the disagreement on these captures is roughly thirty times larger, so whatever separates them from the training data matters more than the decoder. One hypothesis this raised — that the missing colour signal of Section 7.4 is an extraction failure — was tested and **not supported**: among the 12 patients whose captures the architectures agree on best (mean IoU 0.70 or more), the red correlation is −0.37 (p 0.24), against +0.35 (p 0.22) among the other 14. Annotating a subset of these captures is the way to turn disagreement into measured error.

8.5 White balance inside the extractor

Segmenter training read raw image–mask pairs, while the serving path white-balanced each frame before segmenting it, so every learned extractor had been trained on raw pixels and served balanced ones. The masks one model draws on the raw and the balanced version of the same photograph agree at only IoU 0.57 to 0.67 across the three architectures on the 52 captures, and balancing shrinks the mask by about a third. To measure the cost, and to ask whether white balance helps at all, Attention U-Net was also trained with white balance applied, identical in every other respect, and each model was scored under both serving conditions on the same 129 held-out subjects:

Trained on / served	Dice	IoU	Precision	Recall
raw / raw (matched)	**0.8476**	**0.7427**	0.8320	0.8737
raw / gray-world (the pipeline as it was)	0.8360	0.7274	**0.8444**	0.8391
gray-world / raw	0.8329	0.7213	0.7885	**0.8971**
gray-world / gray-world (matched)	0.8407	0.7322	0.8165	0.8789

White balance does not help the extractor: matched raw beats matched gray-world by +0.0069 Dice (95% CI +0.0012 to +0.0118, p = 0.0015). **The mismatch cost +0.0116** (95% CI +0.0066 to +0.0167, $p < 10^{-4}$), more than the change of architecture gained. The two mismatched conditions fail in opposite directions: a raw-trained model served balanced images draws a smaller, more conservative mask, and a balance-trained model served raw images spills. Taken with Section 7.7, which found gray-world costs accuracy when colour is read, no stage of the pipeline has been measured where it helps. Each extractor now records the condition it was trained under and the serving path matches it. This changed deployed masks, so figures measured through the serving path before the correction, including Section 7.5, describe the pipeline as it then was.

8.6 Selection, shipping and the regressor test

Selection follows the standard use of a split: candidates are trained on the training subjects, each one's epoch chosen on the validation subjects, and each scored once on the test subjects. The winner is then retrained on every subject for the epoch count selection found, since nothing is left to stop early on. The shipped weights therefore have no held-out score of their own; the figure quoted for them is the selection score of their architecture. A leaderboard records a fingerprint of the split, so a later architecture can be added without retraining the others, and is refused if the split has changed.

Two production extractors were trained on all 648 images: Attention U-Net on raw pixels (36 epochs), and the same architecture with white balance (23 epochs), each at the epoch count its own selection run chose. Segmentation settles which matches the annotators better; whether white balance changes *haemoglobin* error depends on the regressor and the hospital camera. A tool extracts both sets of crops from any folder of photographs, with crop colour following each extractor's training condition and a manifest of the photographs both accept, and a written protocol specifies the comparison on the regressor: the same photographs, patient-grouped folds, identical seeds and settings, colour augmentation disabled, and a paired per-patient comparison. The deployed U-Net has not been replaced; doing so invalidates every cached crop and is left as a deliberate decision.

9. Verification status

9.1 Verified

- All pipeline stages execute end to end; 50 of 50 automated tests pass, including the HTTP contract the mobile application depends on.
- The service answers on the path the mobile application calls, returns every field the application reads, and returns a retake prompt for captures that fail quality control.
- The learned extractor is trained, tested for bundle integrity, and beats the classical extractor on 217 subjects, including 116 photographs it never saw in training (Section 7.2).
- The Eyes-Defy-Anemia loader is verified against the real download: 217 patients, all ages and sexes parsed, including 19 haemoglobin values stored as text with a decimal comma.
- The colour probe beats its baseline on unseen patients: held-out MAE 1.505 against 1.965 on 190 patients (Section 7.5), measured before the correction in Section 8.5.
- Attention U-Net segments better than the U-Net on 129 held-out subjects: Dice 0.8476 against 0.8372, paired difference +0.0105 (95% CI +0.003 to +0.020, p = 0.007) (Section 8.3).
- Each extractor is served the kind of image it was trained on, with automated tests covering every combination (Section 8.5).
- Quality control enforces every statistic it measures, including illuminant neutrality and clipping.

- The per-patient WHO threshold is validated against real laboratory values: a fixed single cutoff misclassifies 15% of the 26-subject set, all as false negatives (Section 4.4).
- A three-category screening output, with the category probability taken from the model's own errors, catches 93% of anaemic patients while leaving 12% undecided, under a protocol that tunes nothing on the patients being scored (Section 7.8).

9.2 Measured and found wanting

- The classical extractor's masks extend onto sclera: median redness 9.6 against 16.0 for the learned extractor on the same pixels, and held-out R^2 0.074 on unseen external photographs.
- The 26-subject set shows no physiological colour signal under either extractor, and eyelid eversion, which varies widely between its captures, changes measured colour (Section 7.4).
- No model tested transfers between the two cohorts' cameras better than roughly baseline (Section 7.6).
- Predictions shrink toward the cohort mean, weakening classification near the WHO thresholds. This under-flags whichever group's average lies above its own cutoff — anaemic women were missed 17% of the time against 5% for men, corrected to 4% and 12% by drawing the category cut per sex (Section 7.8).
- Tissue texture and vessel-like features add nothing once site is controlled for: an apparent R^2 gain of 0.024 pooled falls to –0.021 correlation (p 0.76) after controls, and is not adopted (Section 7.9).
- Widening the extractor's mask lowers accuracy monotonically across eight variants, so the signal is not being clipped by too tight a mask (Section 7.9).
- Gray-world white balance lowers segmentation accuracy: matched raw beats matched gray-world by 0.0069 Dice, and serving a raw-trained segmenter balanced images cost 0.0116 (Section 8.5).
- On the 26-subject captures the three architectures agree at only IoU 0.648, far below what their held-out scores suggest (Section 8.4).
- The blur threshold looks slightly too aggressive for real captures.

9.3 Not yet established

- Accuracy on the hospital's own camera. No figure in this report applies to it without recalibration (Section 7.6).
- Whether lower-haemoglobin children were everted harder: suggestive (ρ –0.46) but not significant under fully blind rating (Section 7.4).
- Whether the deployed extractor's loss of self-consistency under zoom and blur is real or run-to-run variance; a second training seed is needed.
- The CP-AnemiC loader has not been checked against a real download.
- The clinical targets the screening categories should meet: what share of anaemic patients must be caught, and how large an undecided group is acceptable. The 90% used throughout Section 7.8 is an example for testing, not an agreed target.
- Whether a model trained end to end reads the conjunctiva or the photograph around it; the site-held-out and masked-input evaluations in Section 7.9 are not yet run on a network.
- Whether white balance changes haemoglobin error from the regressor. The comparison is specified and the two extractors it needs are trained (Section 8.6).
- Whether the one-point Dice margin between architectures exceeds run-to-run variance; each was trained with a single seed.
- How wrong the extractors are on captures unlike their training data; the 26-subject captures have no annotations (Section 8.4).
- No clinical validation of any kind has been performed.

10. Data

Four public datasets are in use or identified. The professor has arranged hospital data, which will be the primary source. The image data is not committed to the repository; `DATASETS.md` records sources, licences, download commands and the manifests needed to reconstruct every training set.

Source	Images	Hb	Masks	Role
Eyes-Defy-Anemia	217	Yes	101 usable	Trains and tests the extractor (101 masks); 217 patients with haemoglobin for the colour probe
Eye Conjunctiva Segmentation Dataset	547	No	Yes, ×2	Smartphone captures with two annotators; trains and tests the extractor
CP-AnemiC	710	Yes	No	Not yet downloaded; regression only
26-subject set	52	Yes	No	26 children, both eyes, full blood count; source to be cited. Hand-cropped exports; no colour signal, and eyelid eversion varies between captures (Section 7.4)

Results are reported per cohort and not pooled: the cohorts differ in camera and population, their colour features are several standard deviations apart, and in the 26-subject set the direction of the relationship differs. The 26-subject set carries a full blood count alongside haemoglobin; RDW and MCV distinguish iron-deficiency anaemia from other types, a possible later extension. It is a publicly available dataset rather than one collected for this project, but its original source and licence are not yet recorded here; both must be cited before any figure derived from it appears in a published write-up.

11. Risks

Risk	Mitigation
The model does not transfer to the hospital's camera	Measured already (Section 7.6); plan for per-site recalibration or a colour reference in frame
A model fits a cohort through something other than haemoglobin	A physiological check on every fitted equation, and every result compared across two independent cohorts (Section 7.4 was found this way)
Hospital data arrives in an unanticipated format	Loaders are isolated behind a common record type; an inspection script verifies a dataset before training
A model is served images unlike those it was trained on	Each extractor records its training condition, serving matches it, and tests cover every combination (Section 8.5)
Results overstated in the write-up	Baselines, permutation tests, grouped splits and held-out reporting are built into the pipeline rather than left to discipline

12. Next steps

#	Step	Outcome
1	Run the white-balance comparison on the hospital data with the ResNet regressor, using the two production extractors and the written protocol	Whether white balance changes haemoglobin error, which segmentation alone cannot say
2	Annotate at least 15 of the 26-subject captures	Turns disagreement between architectures into measured error on captures unlike the training data (Section 8.4)
3	Train each architecture with a second seed	Whether the one-point Dice margin, and the U-Net's robustness loss, exceed run-to-run variance
4	Decide whether to replace the deployed U-Net with Attention U-Net trained on raw pixels	Better masks; invalidates cached crops and downstream features
5	Standardise capture: fixed gentle eversion, consistent framing, original camera files, and time since blood draw recorded	Removes the confound found in Section 7.4
6	Discuss a printed colour reference in the capture protocol	Camera-independent colour, if adopted
7	Record the source and licence of the 26-subject dataset	Required before any figure derived from it is published
8	Download CP-AnemiC and verify its loader	Up to 710 further labelled images

Method record. Every approach tried, including those rejected, is recorded with its reason in `RESEARCH_LOG.md` and `WORKLOG.md`, with each experiment reproducible from `experiments/`.

Assessment. Extraction is in good shape, and its choice now rests on the right evidence: evaluated as segmentation on held-out subjects, Attention U-Net improves on the U-Net, and correcting a train–serve white-balance mismatch improved it by more again. The colour probe established that conjunctival colour tracks laboratory haemoglobin at scale. The open problems are portability and capture: colour measured by one camera does not transfer to another, the extractors disagree on captures unlike their training data, and a third public dataset shows no colour signal, in part because eyelid eversion changes conjunctival colour. None is solved by more data from the same cameras; all need captures taken under a fixed protocol on the camera that will be used, which is what the hospital data provides.